

Triangles, tension and compression

Name: _____

Class: _____

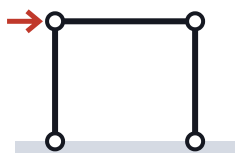
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Tension: a beam being pulled (stretched). **Compression:** a beam being squashed.

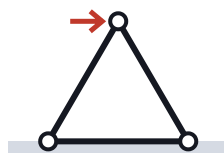
A triangle can't change shape unless one of its sides changes length. 10 kN is about the weight of 1 tonne.

Part A · Structures

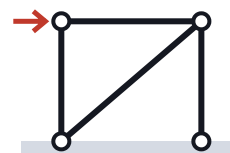
A1 Each frame is made of beams joined by pins that can turn. Which frames keep their shape when you push sideways on the top? Explain. [2]



(a)



(b)



(c)

A2 Is each of these being pulled (tension) or squashed (compression)?

- a) the rope holding up a swing b) a table leg c) the cables of a suspension bridge d) the top beam of a truss bridge [2]

A3 Vehicle weights.

- a) The lorry weighs 80 kN. About how many tonnes is that? b) The crane weighs 100 kN. How many tonnes? [2]

A4 A bridge uses 4 pieces of road at £15 each, 6 straight wooden beams at £8 each and 4 diagonal wooden beams at £11 each. What does it cost? [2]

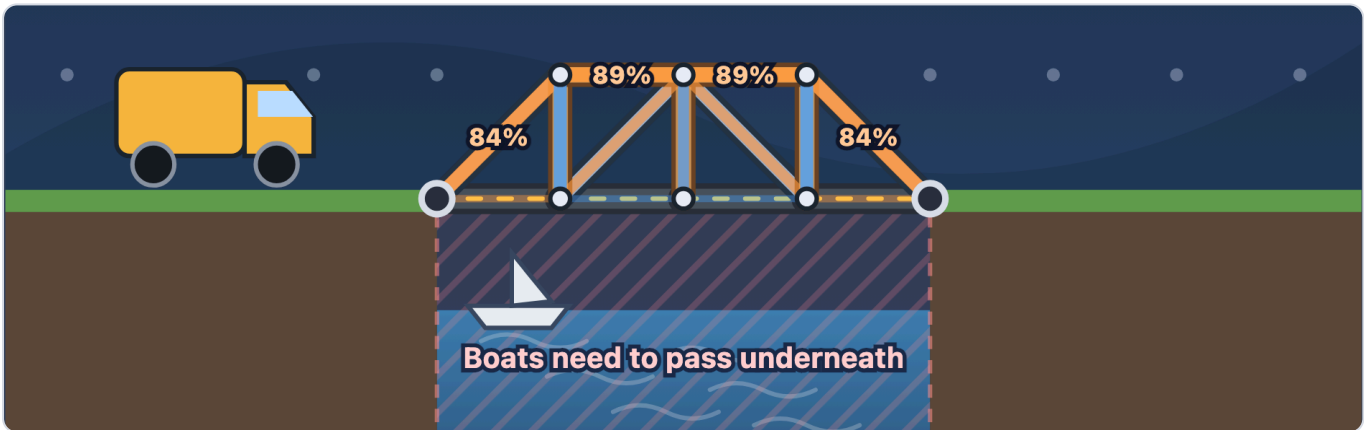
Part B • Play levels 1 to 4

Tick **Show forces**. Before you drive, predict which beam will work hardest. Record your best design for each level.

Level	What I built	Cost	Stars	What went wrong, and how I fixed it
1		£		
2		£		
3		£		
4		£		

Part C • Read the forces

This is **Push and Pull** with **Show forces** ticked: **blue** beams are pulled, **orange** beams are squashed. The percentages show how close the hardest-working beams come to their limit as the truck crosses. The four diagonal beams are steel; the top beams and the uprights are wood.



C1 Are the beams along the top pulled or squashed? And the upright beams? [2]

C2 The top beams show 89%. What does that mean? [1]

C3 The steel end diagonals are squashed with a force of about 76 kN. A wooden diagonal can only take 30 kN of squashing. Why is steel used there? [2]

C4 Steel costs twice as much as wood. Why not make every beam steel? [2]

Part D • Explain

D1 Triangles are used in cranes, roof trusses, electricity pylons and bicycle frames. Explain why engineers use them. [3]

Forces in structures

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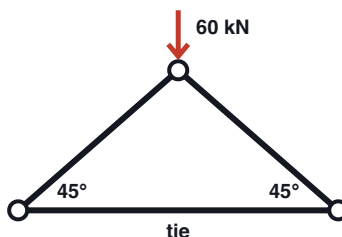
Force F at angle θ to the horizontal: horizontal part $F \cos \theta$, vertical part $F \sin \theta$. $\sin 45^\circ = \cos 45^\circ \approx 0.707$

Buckling: a squashed beam's strength is proportional to $1 \div \text{length}^2$.

Load W at the middle of a span L : biggest bending moment $M = W L \div 4$. Force in each chord $\approx M \div \text{depth}$.

Wood: pulls 70 kN, squashes 60 kN (30 kN diagonal). Steel: pulls 200 kN, squashes 180 kN (90 kN diagonal).

1 Resolving forces. Two beams at 45° meet at the top and are tied together at the bottom. A load of 60 kN pushes down on the top joint. Ignore the beams' own weight.

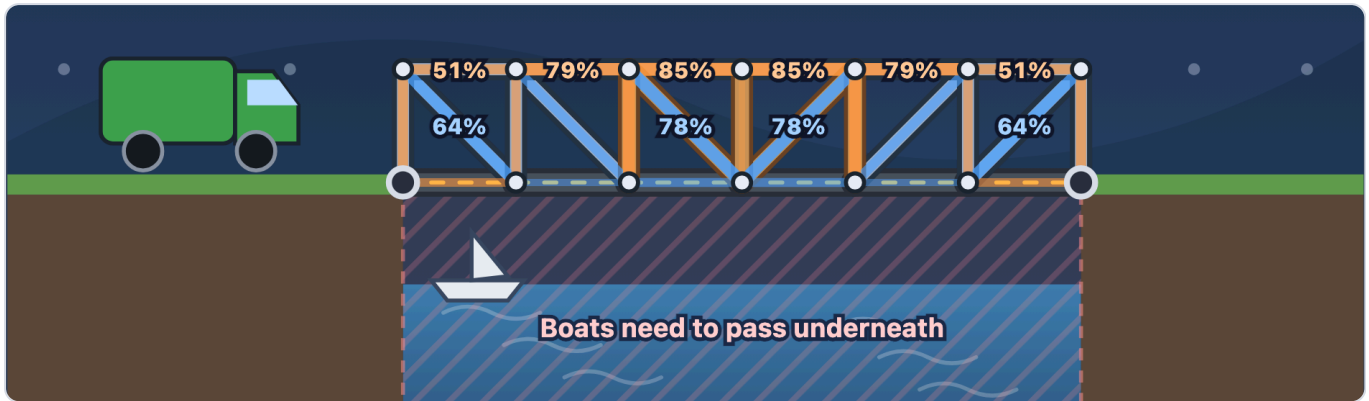


a) Each sloping beam carries half the load vertically. Show that the force in each sloping beam is about 42 kN. Is it pulled or squashed? [3]

b) What is the force in the tie at the bottom? Is it pulled or squashed? [2]

c) The sloping beams are diagonals in Bridge Builder. Should they be wood or steel? Should the tie be wood or steel? [2]

2 Buckling (Buckle Up). This Pratt truss carries an 80 kN lorry. Blue beams are pulled; orange beams are squashed.



a) A straight wooden beam (1 square long) buckles at 60 kN. A diagonal is $\sqrt{2}$ squares long. Show that it buckles at 30 kN. [2]

b) At what force would a wooden beam 2 squares long buckle? [1]

c) In this truss the long diagonals are pulled and the short uprights are squashed. Explain why that is a good design. [2]

3 Go Deeper. An 80 kN lorry stands in the middle of a 32 m span. Ignore the bridge's own weight.

a) Calculate the biggest bending moment. [1]

b) Calculate the force in the top chord if the truss is 4 m deep, and if it is 8 m deep. [2]

c) For each depth, could straight wooden beams carry the top chord force? Could steel? [2]

d) Explain why long bridges are built deeper in the middle. [2]

4 Cable-Stayed. A straight cable runs from a tower top down to the road and holds up 20 kN of the road's weight.

a) Calculate the tension in the cable if it is at 45° to the road, at 30° , and at 15° . [3]

b) Use your answers to explain why cable-stayed bridges have tall towers.

[2]

5 Materials and cost. Wood pulls 70 kN for £8 a square; steel pulls 200 kN for £16.

a) Which gives more pulling strength per pound? Show your working.

[2]

b) The cheapest designs still use plenty of wood. Suggest two reasons why. [2]

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[2]

6 Evaluate. A 100 kN crane must cross a 40 m river. Evaluate three designs: a deep truss, a cable-stayed bridge, and a bridge with a support in the middle of the river. Use evidence from Bridge Builder.

[6]

Continue on lined paper if you need more space.